3.8 Remembering Things

The chapter's capstone

WHAT THIS LECTURE DOES

Takes the raw file I/O from 3.7 and wraps it in functions, then uses them properly:

1. **Load** whatever friends are already saved
2. **Show** them
3. Let the user **add more**
4. **Save** the combined list back

The friends list grows a little more every time you run the program.

WRAPPING FILE I/O IN A FUNCTION

```
std::vector<std::string> loadFriendsFromFile(const std::string& fileName) {
    std::vector<std::string> friends;
    std::ifstream inputFile(fileName);

    std::string friendName;
    while (std::getline(inputFile, friendName)) {
        friends.push_back(friendName);
    }

    return friends;
}
```

THE OTHER HALF

```
void saveFriendsToFile(const std::vector<std::string>& friends,
                       const std::string& fileName) {
    std::ofstream outputFile(fileName);
    for (const std::string& friendName : friends) {
        outputFile << friendName << "\n";
    }
    std::println("Saved {} friend(s) to {}", friends.size(), fileName);
}
```

LOAD FIRST, THEN COLLECT MORE

```
std::vector<std::string> friends = loadFriendsFromFile(fileName);

std::println("\nYou already have {} friend(s) saved:", friends.size());
for (const std::string& friendName : friends) {
    std::println(" - {}", friendName);
}

std::vector<std::string> newFriends = collectMoreFriendNames();
friends.insert(friends.end(), newFriends.begin(), newFriends.end());

saveFriendsToFile(friends, fileName);
```

PROOF IT PERSISTS

First run

`friends.txt` doesn't exist yet

```
You already have 0
friend(s) saved:
```

Second run

picks up right where you left off

```
You already have 2
friend(s) saved:
 - Grace
 - Hopper
```

WHAT THIS PROGRAM ACTUALLY NEEDED

NEED	C++ BUILDING BLOCK
Say something to the screen	Output
Do a named, reusable action	A function
Ask the user something	Input
Compute something new	Arithmetic
Follow a different path	`if` / `else`
Repeat until told to stop	A loop
Hold a growing list	`std::vector`
Remember across runs	File I/O

WHAT C++'S LATEST FEATURES BOUGHT US

- `std::print` / `std::println` - a single `{}`-style format string instead of chained `std::cout <<` - readable from lecture one
- **Brace initialization (`{}`)** - `int age{};` refuses to silently truncate a value you didn't mean to lose - the same habit chapter 4 explains the **why** of

> None of this was necessary to teach the underlying ideas - but it made the code you typed along the way noticeably less ceremonial.

End of Chapter 3

First Steps

Next: Chapter 4 — Variables and Data Types